

Recent advances in the neuroscience of spontaneous and off-task thought: implications for mental health

Received: 27 May 2022

Accepted: 25 August 2023

Published online: 2 November 2023

 Check for updates

Aaron Kucyi¹✉, Julia W. Y. Kam², Jessica R. Andrews-Hanna³, Kalina Christoff⁴ & Susan Whitfield-Gabrieli⁵

People spend a remarkable 30–50% of their awake life thinking about something other than what they are currently doing. These experiences of being ‘off-task’ can be described as spontaneous thought when mental dynamics are relatively flexible. Here we review recent neuroscience developments in this area and consider implications for mental well-being and illness. We provide updated overviews of the roles of the default mode network and large-scale network dynamics, and we discuss emerging candidate mechanisms involving hippocampal memory (sharp-wave ripples, replay) and neuromodulatory (noradrenergic and serotonergic) systems. We explore how distinct brain states can be associated with or give rise to adaptive and maladaptive forms of thought linked to distinguishable mental health outcomes. We conclude by outlining new directions in the neuroscience of spontaneous and off-task thought that may clarify mechanisms, lead to personalized biomarkers and facilitate therapy developments toward the goals of better understanding and improving mental health.

Spontaneous thought and its links to mental health

The past several decades have brought a wealth of research into the neural underpinnings of processes evoked by experimentally directed tasks. However, typical adults spend much of their waking life entertaining thoughts that extend beyond the task at hand (termed ‘off-task thought’) and/or that emerge spontaneously, relatively free from constraints that guide and stabilize cognition. For example, depending on how these types of experience are sampled, being off-task has been estimated to occupy a striking 30–50% of daily time awake in adults^{1–5}.

Having spontaneous and off-task thoughts is a normal, and typically healthy, part of everyday life that can involve reminiscing,

reflecting, imagining, fantasizing, problem-solving or generating creative ideas^{6–8}. Yet such experiences—to which an enormous amount of daily time is dedicated—can also take on forms that are negative in character. For example, rather than reminiscing about the past, an individual may repetitively dwell on previous negative experiences; instead of adaptively planning for the future, one may engage in uncontrollable worry or suicidal ideation. With increasing prevalence rates of psychiatric symptoms and illness in recent years⁹, it is likely that dysfunctional manifestations of inner experience have been increasing in frequency. Given that a proportion of psychiatric patients are resistant to currently available treatments¹⁰, there is a need for novel therapies, such as neuromodulation, to correct disruptive patterns of thought. Thus, understanding neural mechanisms of spontaneous and

¹Department of Psychological and Brain Sciences, Drexel University, Philadelphia, PA, USA. ²Department of Psychology and Hotchkiss Brain Institute, University of Calgary, Calgary, Alberta, Canada. ³Department of Psychology, Cognitive Science, University of Arizona, Tucson, AZ, USA. ⁴Department of Psychology, University of British Columbia, Vancouver, British Columbia, Canada. ⁵Department of Psychology, Northeastern University, Boston, MA, USA. ✉e-mail: aaron.kucyi@drexel.edu

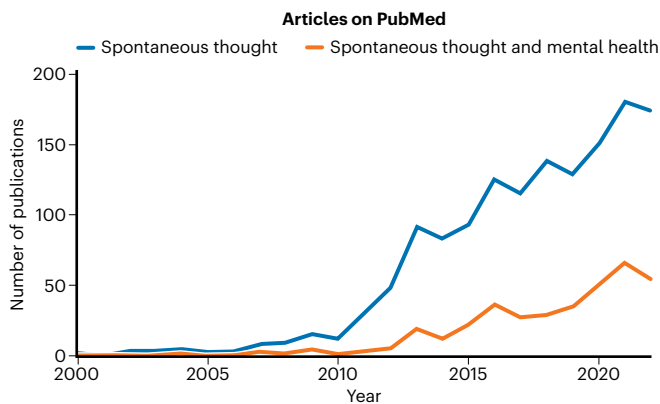


Fig. 1 | Number of publications on spontaneous thought and mental health over the past two decades. PubMed search query for spontaneous thought: ('spontaneous thought' or 'mind wandering' or 'task-unrelated thought' or 'stimulus-independent thought' or 'freely moving thought' or 'self-generated thought'). PubMed search query for mental health: ('mental health' or 'positive affect' or 'negative affect' or 'emotion' or 'mood' or 'sadness' or 'happiness' or 'distress' or 'emotional state' or 'depression' or 'anxiety' or 'ADHD' or 'schizophrenia').

off-task thought is of increasing societal and clinical importance^{11–13}, an urgency mirrored by growing scientific interest^{14–18} (Fig. 1).

Advances in our understanding of spontaneous thought have linked it to dynamic patterns of brain activity that fluctuate across various timescales^{14,15}. The recently proposed dynamic framework of thought (DFT)^{14,19} describes spontaneous thought as emerging in the course of interactions across large-scale brain networks such as the default mode network (DMN), with its core and medial temporal lobe (MTL) subcomponents, as well as the salience and frontoparietal control networks. Mapping the dynamic interactions across large-scale brain networks to the rich varieties of covert, subjective experience presents some unique scientific challenges that have led to novel developments in experimental paradigms. The neural basis of spontaneous thought, along with off-task thought, has so far largely been investigated using functional neuroimaging^{14,15}, but the field is rapidly maturing, incorporating insights from causal experiments (neurostimulation and lesion studies)²⁰, neurophysiological mechanisms of memory^{16,21} and dynamics of ascending neuromodulator systems^{16,18,22}.

In this Perspective, we cover the most recent developments in methodologies and advances in the study of the neural bases of spontaneous and off-task thought with an emphasis on their implications for mental health. We provide an updated overview of the DMN and broader, large-scale networks. Beyond those networks, we discuss candidate mechanisms that involve spontaneous activity in hippocampal memory and ascending neuromodulatory systems. Throughout, we explore how distinct brain states can be associated with and give rise to various forms of healthy and unhealthy thinking patterns. Finally, we outline new directions in the neuroscience of spontaneous and off-task thought that may considerably advance mental health research and clinical practice.

Conceptual and methodological considerations

Before reviewing the neural bases of spontaneous and off-task thought, we consider the conceptual and methodological issues that provide background for the interpretation of research findings. We cover some definitional issues surrounding the evolving concept of 'spontaneous thought', examine the evidence for how the related but distinct construct of off-task thought relates to mental health and review some of the challenges when it comes to investigating spontaneous thought along with the emerging techniques that are beginning to help us overcome these challenges.

The concept of spontaneous thought is still evolving

The term 'spontaneous thought' has become prominent in the scientific literature on a broad range of inner experiences such as mind wandering and dreaming²³. However, notions of what spontaneous thought is exactly and how it is best defined continue to evolve. One important outstanding question remains on whether certain forms of repetitive thought (for example, rumination and worry) should be viewed as spontaneous thought^{24,25} or, as the DFT argues, are better characterized as a distinct mode of thought that unfolds in a highly constrained, inflexible manner¹⁴. There are also broader philosophical concerns regarding the spontaneity of thought²⁶. One proposed way to determine spontaneity is to enquire whether the experiencer could identify a cue or trigger²⁷. However, even in a context where a thought is not initiated consciously by an external stimulus or explicit cognitive control, factors that are typically unobserved, such as interoceptive inputs, may trigger experiences²⁸. Similar issues have been discussed regarding spontaneity in brain activity²⁹.

Beyond considering how mental activity is initiated, it is also important to account for the dynamic processes that ensure the continuity of thought³⁰. The DFT^{14,19} offers an explicit definition of spontaneous thought that hinges on a crucial distinction made by the DFT between mental contents and dynamic thought processes. Contents describe what someone is thinking about, including dimensions such as relevance to the current performance of a task, affective valence (for example, feels positive or negative), temporal orientation (for example, past or future-oriented) and sensory imagery (for example, mentation in the visual or auditory modality). By contrast, dynamics describe how thoughts change from one mental state to another. Within the DFT, spontaneous thought occurs when those dynamics are relatively flexible and thoughts can change easily and freely (rather than being limited to repetitively recurring content). Conversely, according to the DFT, the spontaneity of thought is reduced when its flow is deliberately constrained (that is, by cognitive control) or automatically constrained (that is, by perceptual or affective salience mechanisms that lead to attentional capture). Thus, within the DFT, spontaneous thought is defined based on the dynamics of thought rather than its content, and mind wandering is defined as a form of spontaneous thought that unfolds relatively freely, with low deliberate and automatic constraints¹⁴. Dreaming and creative thinking are also defined as forms of spontaneous thought with distinct deliberate and automatic constraint profiles.

Despite the emphasis on thought dynamics within the DFT, the vast majority of empirical investigations have focused on measuring thought contents rather than dynamics³¹. Mind wandering, specifically, has frequently been operationalized based on thought contents, although recent empirical investigations have begun to measure subjective, dynamic features such as the extent to which thoughts shift around freely^{31,32}. Alternatively, thought dynamics have been characterized by quantifying how experiences change with different durations of time passing³³.

Acknowledging ongoing philosophical debates and outstanding conceptual issues (see refs. 25,34 for detailed discussions), this article reviews a variety of mental experiences and is inclusive of off-task thoughts, experiences described by the DFT as spontaneous thought and thoughts that arise and unfold in an automatically constrained or habitual manner. All of these classes of thought may be indirectly elicited by environmental cues in everyday life (for example, by initiating mental associations)^{35,36}, but they are not, by definition, directly initiated by explicit cues that instruct people to deliberately engage in specific thoughts. Shared features across classes are a focus on inner mental content (as opposed to immediate physical stimuli) and the involuntary nature of thought unfolding. According to the DFT, the category of automatically or habitually constrained thought is of particular interest because it includes phenomena such as rumination and obsessive thought, which are of clinical significance¹⁴. However,

automatic constraints on thought and their clinical implications remain relatively understudied. Instead, most empirical investigations have focused on the related but distinct phenomenon of off-task (or task-unrelated) thought, which we turn to in the next section.

On the relationship between off-task thought and mental health

Research has focused overwhelmingly on deleterious cognitive and health effects of thoughts that are unrelated to the task at hand (that is, off-task thoughts that can be spontaneous or deliberate). Studies suggest that off-task thought impairs cognitive performance in various experimental and real-world settings, including in executive-control tasks³⁷, sustained attention³⁸, memory-encoding tasks³⁹, in classroom lectures⁴⁰, during reading⁴¹ and during automobile driving⁴². Off-task thought can lead to perseverative thinking such as rumination and worry^{43,44}, lowered mood^{2,45}, increased stress⁴⁶ and impaired sleep⁴⁷. Excessive off-task thought—particularly when initiated unintentionally—has been associated with symptoms of depression⁴⁸, obsessive-compulsive disorder⁴⁴, eating disorders⁴⁹ and attention deficit hyperactivity disorder (ADHD)⁵⁰ in certain task contexts⁵¹.

By contrast, positive effects on mental well-being often become apparent when evaluating characteristics beyond the sheer presence of off-task thought, such as what people think about and how thought unfolds⁸. For example, although off-task thoughts are often associated with less positive effect than on-task thoughts, off-task thoughts are still, on average, experienced as mildly pleasant, positive and enjoyable⁵². In some situations, the content of off-task thought may lead to subsequent improvements in mood^{53,54}. Whereas negative mood has been associated with thinking about past events^{55–57}, people tend to daydream more frequently about the future than the past⁵⁸. This ‘prospective bias’ of thought can be adaptive, for example, benefiting problem-solving and creative achievement⁶ or advancing social relationships⁵⁹ (although positive content in future-oriented off-task thought is reduced in individuals with depression symptoms⁶⁰). Past-related thoughts can also be adaptive, promoting meaningful connections with the current and future self⁶¹. Engaging in off-task thought may promote insight—moments when one becomes abruptly aware of a solution to a problem, often associated with a pleasurable feeling of reward⁶². In addition to thought content, there is increasing recognition for the mental health relevance of dynamic dimensions. For example, thought described as freely moving (that is, unconstrained) has been associated with momentary positive affect^{32,63}.

Context is another major factor that determines how off-task thought interacts with mental health^{8,17}. For example, the content and dynamics of off-task thought depend on whether a task is high or low demand⁶⁴ and whether thoughts are sampled in the laboratory or in the real world⁶⁵. In one study, several patterns of off-task thought were identified based on distinct content, and among these patterns, emotional thoughts showed the least generalizability between laboratory and real-world environments across individuals⁶⁵. Overall, how off-task thought interacts with mental health is highly dependent on factors such as thought content, dynamics and context. Importantly, there is tremendous variability between—and even within—individuals that is critical to consider when assessing impacts on mental health.

How can spontaneous thought be studied in the brain?

The study of spontaneous thought in the brain faces a unique challenge: how can mental activity that is initiated without any instructive, external trigger—and that is unpredicted even by the experiencer themselves—be measured or quantified and attributed to brain events? Whereas typical experimental models in cognitive neuroscience employ structured tasks or stimuli to trigger brain responses that are time-locked to external events, distinct approaches are needed to generate insights into the neural basis of spontaneous thought.

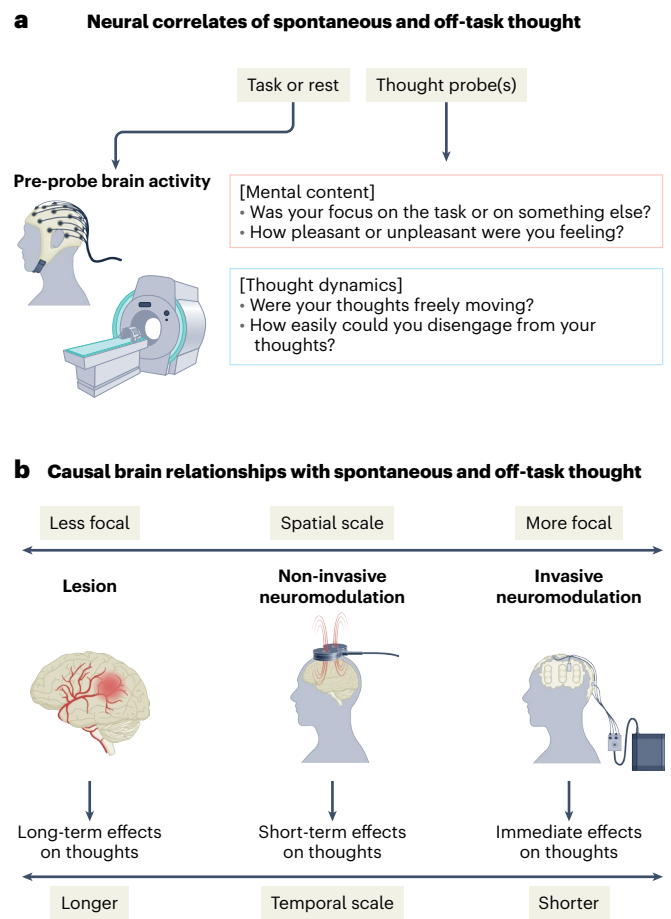


Fig. 2 | Linking spontaneous thought and off-task thought phenomenology to neural measures with experience sampling. **a**, Schematic example of the use of online experience sampling to examine neural correlates and predictors of spontaneous and off-task thought. **b**, Distinct techniques for examining causal brain relationships with spontaneous thought that provide insights at multiple spatial and temporal scales. Figure created with [BioRender.com](https://www.biorender.com).

Before the past two decades, hypotheses about neural mechanisms were limited largely to interpretive comments when authors observed patterns of elevated brain activity during task-free, waking rest, such as the electroencephalography (EEG) posterior alpha wave (reviewed in ref. 66)^{67,68}. However, ‘resting state’ models alone cannot generate direct insights into the brain’s representation of subjective experiences during spontaneous thought, which require that brain activity be linked directly to self-reports of such experiences^{14,15,69}. Starting in the 1990s, positron emission tomography (PET)^{68,70} and functional magnetic resonance imaging (fMRI)^{71,72} studies began to examine relationships between brain activity and the self-reported frequency of off-task thought, which participants estimated retrospectively after completing brain scans.

More recently, online experience sampling during neuroimaging has been applied to examine the finer temporal dynamics within individuals⁷³ (Fig. 2a) (see also ref. 74 for an earlier EEG example). Here, people are intermittently presented with self-report prompts (‘thought probes’) asking them to rate or describe their most immediate experience. Thought probes must be designed carefully and with the aim of identifying valid and reliable brain-experience relationships, as it has been shown that the specific phrasing and manner of presentation^{4,5,75} as well as the population being studied⁷⁶ can influence participant responses. Despite such issues, experience sampling has

BOX 1

Neural basis of mind blanking and implications for mental health

When people are disengaged from the immediate environment, they are usually engaged in their own thoughts, but there are also moments of mind blanking when no mental contents can be recalled. Although less studied with respect to mental health than mind wandering and spontaneous thought, studies suggest that mind blanking is increased in ADHD¹⁹⁸ and impairs cognitive performance in depression¹⁹⁹. The neural correlates of mind blanking have recently been explored. When people were instructed to intentionally engage in mind blanking, deactivation of the DMN was found (that is, opposite to the typical pattern for off-task thought)²⁰⁰. Online experience sampling with EEG and fMRI has further confirmed that mind blanking has distinct neural correlates compared with mind wandering or off-task thought^{89,90}. These findings should motivate the inclusion of mind-blanking response options in experience-sampling protocols in mental health contexts. Notably, protocols should carefully distinguish between absence of thought versus absence of all conscious experience, as these distinct features may have been conflated in previous studies²⁰¹.

been used successfully to identify brain-experience relationships in groups of participants on the order of seconds to minutes⁷⁷. Alternative approaches to intermittent experience sampling, such as self-caught reporting of experience, may provide temporal insights into the arising of spontaneous thoughts and their neural antecedents (especially in trained individuals such as meditation practitioners)^{78,79}. Moreover, free-association-based thought-sampling⁸⁰ and think-aloud⁸¹ paradigms have been used, albeit more rarely, in combination with neuroimaging. In some cases, verbal or auditory cues have been used to elicit involuntary mental associations, thereby manipulating the occurrence of spontaneous thought^{82–84}.

Researchers have only recently begun to characterize the brain bases of spontaneous thought's more granular features that have relevance to mental health. Multi-dimensional experience sampling¹⁷ during neuroimaging has been used to examine thoughts that vary in terms of emotion⁸⁵, self-relevance²⁸, level of detail⁸⁶, modality and perceptual features of sensory imagery⁸⁷, and dynamic properties⁸⁸. Recent work has also explored the neural basis of 'mind blanking', an absence of reportable thought (Box 1)^{89,90}. Studies involving online experience sampling with brain-activity measures are beginning to emerge in patients diagnosed with ADHD^{91,92} and treatment-resistant depression⁹³. Correlative neuroimaging findings are being complemented with insights from interventions⁹⁴ as well as causal approaches in neuroscience such as transcranial direct current stimulation (tDCS), intracranial electrical stimulation and lesion studies, providing insight into the emergence of spontaneous thought across a wide range of timescales (Fig. 2b). Together, the study of spontaneous thought in the brain is quickly expanding in terms of the models used, phenomenology explored, clinical populations examined and neuroscience techniques applied. With these emerging developments, it is timely to examine their potential to yield new insights, given what is currently known about spontaneous thought in the brain.

Recent neuroscientific findings with relevance to spontaneous thought and their links to mental health

Default mode network: an update

The DMN has received by far the most attention in reference to the neural basis of spontaneous and off-task thought. The DMN has also been studied extensively in relation to mental health and illness. We provide updates on different features of the DMN, including its activation and deactivation profiles, functional connectivity, subsystems and the effects of perturbation. We describe how these features inform the neural basis of spontaneous thought and the implications for mental health.

Activation and deactivation. Regions that comprise the DMN—including large portions of the posteromedial cortex, medial prefrontal cortex and lateral parietal cortex (among others)—were, initially, largely appreciated for showing relative decreases in blood flow during (external) cognitive tasks compared with passive control conditions⁹⁵. The 'default mode' network^{96,97} has since been shown to be recruited across a variety of internally guided processes such as remembering, envisioning the future and social inference⁹⁸—experiences that are common during spontaneous thought^{68,99}. Moreover, the activity of the DMN is often anticorrelated with, or fluctuates independently from, networks involved in externally oriented attention^{100–102}. Such observations led to the hypothesis that the DMN could be "the core brain system associated with spontaneous cognition"¹⁰³. Subsequently, fMRI studies with online experience sampling identified trial-by-trial associations between off-task thought and increased DMN activation (or decreased deactivation), which was replicated across multiple contexts^{73,99,104,105}.

The implications of such findings for mental health immediately became a topic of interest. In particular, researchers explored the idea that DMN overactivation underlies an inability to disengage from unwanted, off-task thoughts. For example, in attention and psychotic disorders, it was suggested that an inability to suppress the DMN could give rise to attention lapses, performance variability and cognitive impairments^{106,107}. In disorders of mood and anxiety, DMN hyperactivation was hypothesized to underlie maladaptive rumination and worry¹⁰⁷. On the other hand, DMN activation may support healthy forms of thought such as creative idea generation and evaluation¹⁰⁸. Overall, findings highlighted that nuanced interpretations are needed when evaluating DMN activation/deactivation as a potential biomarker or treatment target in mental health contexts. Adding complexity, recent experience-sampling fMRI studies have questioned the notion of a simple one-to-one relationship between DMN activation and off-task thought^{86,109,110}, whereas other paradigms have linked DMN activation to the use of working memory to guide task-related behavior^{111,112}. To better clarify the role of the DMN, it may be important to consider other network features.

Functional connectivity. A distinct defining feature of the DMN is the correlated spontaneous activity between its different regions⁹⁷. Such 'functional connectivity' has been shown to persist during resting states, task performance and even sleep and unconscious states. As within-network functional connectivity appears to be largely preserved across cognitive states¹¹³, it may not purely reflect current thoughts¹⁵. Instead, the strength of correlation within intrinsic networks may be shaped by an intricate combination of factors such as previous life experience, genetic traits, current cognitive state and other 'intrinsic' neuronal processes¹¹⁴.

Resting-state fMRI studies have reported correlations between individual off-task thought tendencies and DMN functional connectivity strength^{115,116}. Moreover, DMN connectivity abnormalities have been reported extensively within various populations in which mental health is impacted^{107,117}. On the basis of the widespread idea that DMN activation signifies off-task thought, authors have sometimes attributed connectivity abnormalities to disruptions in spontaneous thought or

rumination that may arise at rest. Interestingly, distinct recurring brain states can be identified at rest¹¹⁸, and the occurrence rate of a DMN-activation state was recently associated with individual differences in post-rest retrospective reports of unintentional intrusive thoughts about the past¹¹⁹. However, to disentangle state (intra-individual) from trait (inter-individual) relationships, it is critical to examine experiential fluctuations within individuals.

To do so, studies have combined online experience sampling with intra-individual analyses of pre-thought probe functional connectivity^{92,105,110,115,120}. So far, these ‘dynamic functional connectivity’¹²¹ studies have focused almost exclusively on off-task thought. A complex set of findings has emerged, where intra-individual off-task thought has been associated with both increased and decreased within-DMN functional connectivity. Potentially reconciling those findings, recent work revealed both positive and negative correlations with off-task thought depending on the precise DMN subregions involved⁹². Taken together with additional evidence linking DMN connectivity fluctuations with ongoing cognition and behavior, it is becoming clear that DMN dynamics reflect a complex combination of state factors (for example, experience/thought, arousal and attentional fluctuations) and traits of individuals^{15,122}. However, as resting-state functional connectivity studies in clinical populations do not typically include measures of online experience, it has been difficult to directly link alterations of spontaneous thought (or other state factors) to functional connectivity abnormalities. Further investigations that combine resting-state paradigms, dynamic functional connectivity approaches and online multi-dimensional experience sampling may have an important role in clarifying the role of disrupted spontaneous thought in DMN connectivity abnormalities observed in mental illness^{15,69}.

Subsystems. While the DMN as a whole may support a domain-general function, there is a wealth of fMRI evidence for specialized subsystems^{98,123–125}. According to one neurocognitive model¹²⁶, a DMN subsystem anchored in the dorsomedial prefrontal cortex may support high-level abstract and verbal forms of cognition such as mentalizing and language comprehension (a ‘mind’s mind’ form of thought), whereas a subsystem anchored in the posterior MTL may support more contextually specific and imagery-based forms of cognition such as episodic memory and episodic future thinking (a mind’s eye form of thought). Recent findings illustrate the individual-level, precise delineation of DMN subsystems based on intrinsic functional connectivity¹²⁷ and activation during deliberate self-generated thoughts with distinct content¹²⁸. A common finding is that subsystems can be distinguished based on coupling with MTL structures including the hippocampus¹²⁹. The MTL-anchored DMN subsystem has also been termed the ‘posterior-medial memory network’¹³⁰ and is consistently implicated in episodic thought about the past and future.

Understanding the specific roles of DMN subsystems in thought content and dynamics is an active area of investigation. Regarding content, it is likely that subsystems are regularly activated spontaneously during the occurrence of similar categories of thought that have been mapped in task-activation studies. For example, a ventromedial prefrontal region that has been implicated in social cognition (based on task-evoked activation) also shows increased activation with off-task thought involving episodic social content (as detected via experience sampling)¹³¹. Indirect support also comes from studies of individual differences showing that functional connectivity of the MTL, but not the dorsomedial prefrontal, subsystem is associated with daydreaming frequency¹¹⁵ and with retrospectively reported thoughts about the past and future⁹⁹. Moreover, a greater cortical thickness of MTL regions within the DMN has been associated with individual tendencies for a higher level of detail in off-task thought (in both laboratory and real-world settings)^{65,132}. In terms of thought dynamics, it has been hypothesized that the MTL subsystem initiates spontaneous thought during conditions when deliberate constraints are weak¹⁴. This idea is

BOX 2

Intracranial neuromodulation of the DMN and spontaneous thought

Intracranial electrical stimulation²⁰² is a neuromodulation technique that offers more focality compared with tDCS and is similar to deep-brain stimulation approaches²⁰³. A short pulse of high-frequency (50–100 Hz) intracranial stimulation, often applied for the clinical purpose of functional mapping in neurosurgical patients, can immediately change subjective experience or behavior. The change may involve a wide range of perceptual, motor, cognitive and affective effects, including spontaneous vivid experiences of past memories²⁰⁴. Recently, intracranial stimulation effects have been functionally mapped to large-scale intrinsic networks. Interestingly, the elicitation rate of changes in subjective experience or behavior was lower in the DMN (~20%) than in any other intrinsic network²⁰⁵. This finding may seem surprising, but it is notable that the context of intracranial stimulation is typically comparable to a resting state, when internally guided cognition may already be high, thus limiting the potential to elicit effects. When DMN stimulation effects are found, they often involve vivid and highly specific phenomenological experiences. In a notable recent example, stimulation of a region proximal to DMN within the posteromedial cortex caused a patient to report a sense of observing their own thoughts, an experience comparable to self-dissociation in neuropsychiatric illness²⁰⁶. Beyond immediate experiential effects, a recent study showed that intracranial DMN stimulation reduced the capacity to generate creative thoughts²⁰⁷. Such findings provide preliminary insights into the potential short- and long-term impacts that invasive, deep-brain stimulation therapies (for example, for treatment-resistant psychiatric illness) may have on spontaneous thought patterns and their associated mental health outcomes.

supported by fMRI findings in mindfulness practitioners, where self-caught spontaneous thoughts were preceded by MTL activation⁷⁸. We discuss further the potential role of the MTL in initiating spontaneous thought in ‘Hippocampal sharp-wave ripples and replay’.

How might DMN subsystems shed light on the role of spontaneous thought in mental health? A promising avenue involves characterizing relationships with phenomenological experiences, beyond off-task thought. A recent fMRI study has provided insight into the role of the MTL subsystem in the self-relevance and affective valence of spontaneous thought⁸⁰, features that are known to be important to long-term health^{2,45,133}. In a free-association thought-sampling model¹³⁴, participants generated conceptual associations that varied in self-relatedness and negative affectivity. Multivoxel pattern analyses revealed that the self-relevance and affective valence of concepts could be predicted from a variety of brain networks, including key contributions of the MTL and other DMN regions⁸⁰. Such findings suggest that DMN subsystems provide insight into the neural bases of spontaneous phenomenological experiences that are closely tied to mental health.

Effects of perturbation. So far we have reviewed neuroimaging studies that offer correlational evidence. Recently, investigators have begun to explore the possible causal role of the DMN based on lesion studies as well as perturbation with tDCS and invasive stimulation of localized DMN regions (Box 2). Beyond being critical to establishing causal

mechanisms of spontaneous thought, these studies have direct implications for neuromodulation therapies in mental health.

Several investigators have applied tDCS to DMN regions and tested the short-term (presumably reversible) effects on the frequency of off-task thought^{135–137}. However, these studies have so far yielded mixed findings that may depend on the targeted region, stimulation parameters, outcome measure and other factors²⁰. Further research is needed to determine whether or not non-invasive DMN neuromodulation can reliably alter the frequency of off-task thought, and whether or not the effects may have clinical utility in conditions such as ADHD, where off-task thought can occur excessively¹³⁸.

Interestingly, both lesion and tDCS studies provide preliminary converging support for the causal importance of the MTL subsystem in the content of spontaneous thought. Patients with lesions to regions within this subsystem (the ventromedial prefrontal cortex¹³⁹ and the hippocampus¹⁴⁰) and with neurodegeneration affecting this subsystem¹⁴¹ report a reduced frequency of spontaneous thoughts about the past and future. In a recent study in healthy adults, tDCS to the posterior inferior parietal lobule, an MTL subsystem region, resulted in reduced frequency of thoughts about the past involving negative affect¹⁴². Given the associations between past-related thought, negative affect and rumination^{55,56}, these findings preliminarily suggest that targeting the MTL subsystem may improve mood via altering the temporal focus of spontaneous thought.

Large-scale network dynamics beyond the DMN

Despite the field's emphasis on the DMN, it is now widely appreciated that multiple other large-scale brain networks support the diverse content and dynamic qualities of spontaneous and off-task thought. In this section, we provide brief updates on networks beyond the DMN, focusing primarily on the frontoparietal control network (FPCN) and the salience network. We then highlight select, novel findings that have mapped network dynamics to features of spontaneous thought that are important to mental health.

The FPCN, a network that interacts closely with both the DMN and externally oriented systems¹⁴³, has repeatedly been implicated in off-task thought^{14,20}. The FPCN is hypothesized to impose deliberate constraints and to cooperate with the DMN to control the course of thought^{14,144}. Recent tDCS studies have shown both increases and decreases in off-task thought after the stimulation of a dorsolateral prefrontal area within the FPCN^{145,146}. Despite diverging results that may have depended on task context²⁰, these studies hint that modulation of the FPCN may alter deliberate constraints on thought, which can be beneficial in contexts where it is desirable to increase cognitive control and reduce off-task thought.

The salience network is often engaged when orienting attention toward salient stimuli or mental events¹⁴⁷ and is hypothesized to impose automatic constraints on thought¹⁴. Cortical areas of the salience network (for example, anterior insula, mid-cingulate cortex) are tightly associated with subcortical arousal and noradrenergic systems^{148,149}. Although the salience network is not usually activated during off-task thought in fMRI studies (however, see ref. 79), future studies are needed to determine its relevance for thought content and dynamics. Resting-state fMRI combined with separately conducted experience sampling in real-world settings has shown that individual tendencies to engage in negative thinking are associated with functional connectivity between the salience network, the FPCN and the DMN^{150,151}. These findings may be consistent with the salience network participating in thoughts with strong automatic constraints and arousal qualities, such as rumination and worry.

Various other large-scale networks appear to be involved in spontaneous thought content and dynamics. Sensory and motor networks have been linked to processes such as internally generated imagery⁸⁷ and perceptual decoupling¹⁵². Within the context of reading, one study revealed that individuals who had more frequent off-task thoughts

displayed reduced resting-state functional connectivity between the DMN and early visual regions (potentially supporting perceptual decoupling during reading)¹⁵³. The dorsal attention network (DAN) may compete with the DMN to support the movement of attention between spontaneous thought and external stimuli¹⁴. Experience sampling across multiple task conditions has revealed that areas of the DAN (in the parietal cortex) are more activated for on-task relative to off-task thought, regardless of context⁶⁴. Subcortical systems that control ascending neuromodulatory signals probably contribute to the regulation and flow of spontaneous thought¹⁶.

Overall, spontaneous and off-task thoughts involve complex, dynamic orchestration of activity across distinct networks at the whole-brain level. These findings have motivated recent applications of data-driven, predictive modeling analyses¹⁵⁴ that take into account a rich variety of brain network features. In recent work⁹², connectome-based predictive modeling¹⁵⁵ was combined with fMRI and online experience sampling. In healthy adults, a pattern of whole-brain functional connectivity predicted trial-by-trial off-task thought and involved interactions within and between virtually all intrinsic networks to some degree (Fig. 3a). A prominent feature was decreased anticorrelation between the DMN and FPCN during off-task thought, a finding in line with research implicating DMN–FPCN interactions in controlling the course of thought¹⁴⁴. A second prominent feature was decreased coupling between the DMN and sensorimotor regions, a process that may underlie perceptual decoupling¹⁵². Importantly, the network pattern also predicted off-task thought in adults with ADHD. These patients reported a heightened frequency of off-task thought and overexpressed the network pattern. As excessive off-task thought is strongly associated with adverse clinical outcomes in ADHD¹³⁸, these findings give clues regarding complex network interactions that could be considered as therapeutic targets.

Most studies examining network dynamics and spontaneous thought have used fMRI, which has limited temporal resolution, indirectly measures brain activity, and is usually limited to short sessions. To overcome these limitations, one study combined experience-sampling with intracranial electroencephalography (iEEG)¹⁵⁶. Patients with semi-chronic, implanted electrodes (within limited cortical and subcortical regions) were asked to rate task-free fluctuations in subjective mood over several days. Based on inter-electrode temporal coherence, multiple intrinsic functional networks were identified involving the amygdala, hippocampus and salience network regions (for example, insula, cingulate) (Fig. 3b). Self-report mood fluctuations were predicted from the variability of coherence in the beta-frequency range (13–30 Hz) within a network involving amygdala–hippocampus coupling¹⁵⁶. This pattern may be indicative of interactions between the salience network (amygdala¹⁴⁹) and DMN (hippocampus¹²⁹). These iEEG findings provide strong neurophysiological evidence for the relevance of network dynamics to fluctuations in subjective experience, highlighting the role of synchronized oscillatory patterns that have been more broadly linked to cognition¹⁵⁷ but rarely explored in relation to spontaneous and off-task thought. Extending the findings, the iEEG experience-sampling approach was recently applied in treatment-resistant depression for the purpose of identifying a deep-brain stimulation target⁹³. This novel direction highlights how the mapping of large-scale network dynamics to spontaneous thought may provide clinical biomarkers to guide treatment.

Hippocampal sharp-wave ripples and replay

It is likely that the field has focused extensively on large-scale brain networks because their function is detected readily using current human neuroscience methods. However, certain brain events that have been observed in non-human animals display properties that may be consistent with the occurrence of spontaneous or off-task thought. One neurophysiological event that is receiving growing attention is the so-called sharp-wave ripple (SWR)^{16,21,158}. The SWR is an electrophysiological

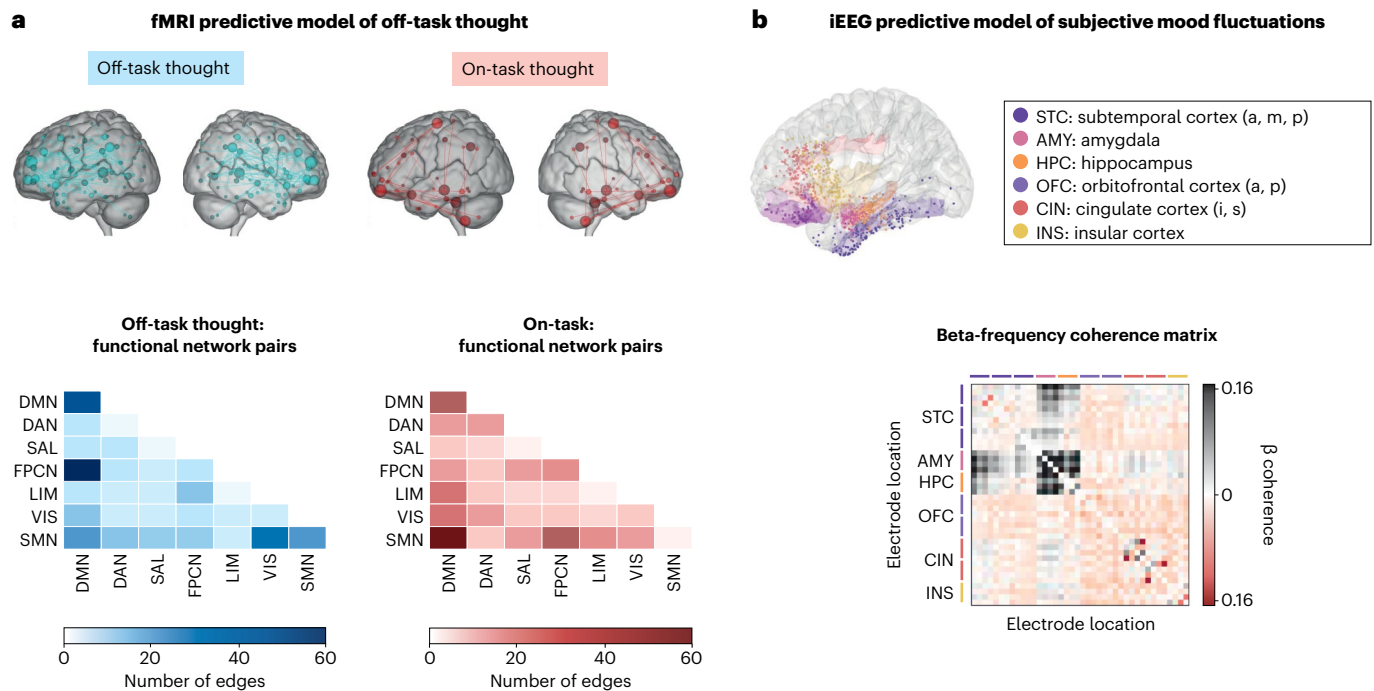


Fig. 3 | Large-scale brain network predictive models of spontaneous fluctuations in experience-sampling ratings. **a**, Top: connectivity features of an fMRI-based functional network model that is predictive of intrasubject off-task thought and on-task attention. Bottom: the number of edges (pairs of regions) between multiple intrinsic networks that contribute to the model. SAL, salience network; LIM, limbic network; VIS, visual network; SMN, sensorimotor network. **b**, Top: implanted electrode locations and neuroanatomical regions

involved in an iEEG-based functional network model that is predictive of subjective mood fluctuations. Bottom: connectivity features (beta-frequency coherence matrix) within a subnetwork involving strong amygdala–hippocampus coupling that is predictive of intrasubject mood ratings. **a**, anterior; **m**, middle; **p**, posterior; **i**, inferior; **s**, superior. Panels adapted with permission from: **a**, ref. 92, Springer Nature Ltd; **b**, ref. 156, Elsevier.

event, generated from highly synchronous neuronal activity, that lasts <150 ms and occurs spontaneously in the hippocampus within the MTL. Information from SWRs is transmitted widely to cortical and subcortical targets. Microscale local field potential signals are typically used to detect hippocampal SWRs, but recently, putative SWRs have been documented at coarser spatial scales in the human brain using iEEG^{159,160} and magnetoencephalography (MEG)¹⁶¹.

There are several reasons to suspect that hippocampal SWRs have a mechanistic role in spontaneous thought. First, SWRs are observed selectively during ‘offline’ periods such as wakeful rest (for example, when animals are remaining still) when external constraints on thought are low¹⁶². Second, SWRs coincide with extrahippocampal neuronal firing sequences that replay past experiences (reverse replay) or construct potential future scenarios (forward replay)¹⁶³. Combinations of these sequences could support unconstrained semantic associations that are characteristic of spontaneous thought²¹. Spontaneous memory reactivations have also been identified in the hippocampus and cortex in human resting-state fMRI¹⁶⁴ and MEG¹⁶¹. Third, hippocampal SWRs^{165,166} and replay events¹⁶⁷ occur in concert with activation of the DMN, the large-scale network most frequently associated with spontaneous thought¹⁶⁸.

Despite these theoretical considerations, a direct link between SWRs and spontaneous thought has not yet been firmly established. Whereas it is possible that SWRs are directly involved in conscious experience, an alternative is that they support preconscious contemplations that influence subsequent experiential phenomena¹⁶². As SWRs have largely been studied in animal models, it has been difficult to draw conclusions about relationships with mental experience. Notably, intracranial electrical stimulation of the hippocampus and nearby white-matter tracts in neurosurgical patients often evokes spontaneous vivid experiences of past events¹⁶⁹. However, it remains unknown whether or not such stimulation influences SWRs. A recent iEEG study

has shown that SWRs increased in frequency before the spontaneous, conscious recall of previously viewed images¹⁶⁰, providing correlative evidence for their involvement in mental experience.

Several authors have speculated that spontaneous hippocampal SWRs are involved in adaptive forms of thought, such as creative thinking and problem-solving^{21,158}. During SWRs, replay-related hippocampal and cortical sequences can generate new combination ‘strings’ of sequences¹⁶¹. When conditions promote spontaneous and off-task thought (for example, at rest), the continuous stitching together of different sequence combinations provides a potential basis for abstract thinking, novel ideas or sudden insights¹⁵⁸.

By contrast, SWRs and associated replay may lead to disruptive forms of thought in conditions affecting mental health. A MEG study showed that patients with schizophrenia, relative to control subjects, showed a reduction in spontaneous neural replay as well as elevated (potentially compensatory) SWR power¹⁷⁰. Such distortions in resting-state brain activity, which may relate to DMN abnormalities in schizophrenia¹⁰⁷, could be associated with experiences such as disorganized or delusional thinking. In addition, it has been hypothesized that disrupted SWRs and replay play a role in disrupted thought in depression and anxiety; sustained SWRs that drive forward and reverse replay, respectively, could be the bases of uncontrollable worrying and ruminative thoughts¹⁷¹. Overall, SWRs may have key roles in both adaptive and maladaptive thought, and there is need to further characterize these events and their interplay with cortical networks during experience.

Ascending neuromodulatory systems

Ascending neuromodulatory systems—such as the cholinergic, noradrenergic and serotonergic systems—originate in brainstem or subcortical nuclei and send broad projections to cortical networks. These systems, and their complex interactions with one another, strongly influence the neural phenomena described here (that is, DMN activity,

large-scale network dynamics, hippocampal SWRs)¹⁷². Ongoing fluctuations within neuromodulatory systems regulate many aspects of behavior and cognition and have a vital role in mental health. Pharmacological agents that target these systems are regularly used in the treatment of mental illness, yet little is known about the role that neuromodulators play in spontaneous and off-task thought. In this section, we discuss two systems where knowledge is beginning to accumulate: the noradrenergic system and the serotonergic system. Other systems (for example, cholinergic or dopaminergic systems) that have been theoretically implicated in spontaneous thought but require empirical study are beyond the scope of this review (see ref. 16 for an overview).

Noradrenergic system. The noradrenergic system originates in the locus coeruleus, projects diffusely to cortical regions and has a major role in the regulation of global brain states to control vigilance and arousal. Tonic and phasic fluctuations in noradrenaline levels, respectively, refer to slow and fast changes that mediate separate functions¹⁷³. The size of the pupil is often used as a proxy for noradrenergic activity level, which is supported by neurophysiological and neurostimulation evidence^{174,175}. However, this relationship between pupil size and noradrenergic activity is somewhat variable over time¹⁷⁶. Other systems (for example, the cholinergic system) are also associated with pupil size¹⁷⁵. As such, findings from studies that examine pupil size in relation to spontaneous thought should be interpreted as preliminary regarding the involvement of the noradrenergic system.

During off-task thought, tonic pupil size was found to be both elevated and reduced in different studies^{120,177,178}. These findings have been interpreted with reference to the inverted U-shaped relationship that noradrenergic activity has with task performance: too much (high arousal) or too little (low arousal) activity can impair behavior¹⁷³. One suggested possibility is that low levels of noradrenaline, associated with low arousal and increased SWRs¹⁷², promote freely moving and transitory thoughts (mind wandering, according to the DFT¹⁴), whereas high levels promote disjointed thoughts with reduced awareness (which has been referred to as mind blanking)^{16,178}. In contrast to tonic pupil size, phasic responses to stimuli have consistently been found to decrease during off-task thought^{120,179} (see, however, ref. 110). This may reflect a disengagement of arousal responses to external stimuli related to perceptual decoupling. However, phasic pupil dilation, which occurs regularly during wakeful rest, could reflect engagement with inner thoughts. A recent iEEG study showed that spontaneous pupil dilations (resembling task-evoked, phasic dilations) were consistently preceded by neuronal population activations in the anterior insula within the salience network¹⁸⁰. A possibility that could be explored is that coupling between the noradrenergic system and the salience network may impose automatic constraints during worrying and ruminative thoughts.

Taken together, pupillometric studies tentatively suggest the involvement of the noradrenergic system in spontaneous thought. These findings have important implications for the use of stimulant medications that are commonly used to treat mental illness (for example, ADHD and depression) and include the noradrenergic system as a central target. Further research may clarify how pharmacological manipulation alters spontaneous and off-task thought to impact mental health.

Serotonergic system. The serotonergic system originates in the raphe nuclei and, like the noradrenergic system, projects diffusely to most cortical regions. This system acts on a diverse set of receptor subtypes, enabling precise neuromodulation that depends on the target neuronal population. In psychiatric practice, the serotonergic system has long been a target of pharmacotherapies that aim to improve mood (for example, selective serotonin reuptake inhibitors). Also acting on the serotonergic system are psychedelic agents—including psilocybin,

lysergic acid diethylamide (LSD) and dimethyltryptamine—that can have profound, short-term effects on subjective experience as well as potentially longer-lasting therapeutic effects for a variety of mental illnesses¹⁸¹. There has been a recent resurgence of scientific and clinical interest in psychedelics. Because psychedelics impact spontaneous thought in an acute and unique manner, we focus here on insights from these agents as a lens into the role of the serotonergic system.

Although psychedelic agents do not act exclusively on the serotonergic system, their agonist action on the serotonin receptor subclass 5-HT_{2A} has a key role in generating experiential effects¹⁸². The psychedelic experience, which may last between ~30 min and 12 h, involves vivid alterations in perception and feelings that take on many variations but often involve hyperassociative and discontinuous thought, visual imagery/hallucination, intense emotion and an altered or diminished sense of self ('ego dissolution')^{183,184}. Given these features, the psychedelic state was recently incorporated into the DFT as a form of spontaneous thought with lower deliberate and automatic constraints than mind wandering, creative thinking and dreaming¹⁹. This characterization situates the psychedelic state as the most unconstrained and least ruminative known version of thought; thus, understanding the role of the serotonergic system in this state provides a potentially unique window into the neural basis of spontaneous thought.

How psychedelic action at the 5-HT_{2A} receptor leads to subjective effects remains unclear and is an active area of investigation¹⁸⁵. Recently, resting-state neuroimaging has been used to examine the acute effects of psilocybin and LSD. Although multiple neural metrics have been examined, which are difficult to compare across studies, one consistent finding is that LSD reduces functional connectivity between the thalamus and sensory cortical regions^{186,187}, a phenomenon that may be associated with perceptual decoupling¹⁵². Psilocybin and LSD have also been shown to decrease functional connectivity within the DMN^{182,184,188}, which may be linked to subjective experiences of creative insights¹⁸⁹. Given the role of the DMN in self-referential thought, this effect has also been attributed to ego dissolution. However, decreased functional connectivity within the salience network¹⁹⁰ and increased coupling between regions rich in 5-HT_{2A} receptors¹⁹¹ have also been associated with individual differences in psilocybin-induced ego dissolution. Critically, these resting-state studies did not examine intra-individual variability in spontaneous experiences within the psychedelic state. Future studies involving psychedelics, neuroimaging and experience sampling may shed light on the mechanistic role of the serotonergic system in functions such as enhancing the spontaneity of thought, reducing rumination and exerting long-term benefits to mental health.

Conclusions and future directions

In this review, we have highlighted some of the diverse approaches and findings emerging from the rapidly growing body of research on the neuroscience of spontaneous and off-task thought. We explored novel findings that build on the well-studied role of the DMN and large-scale network dynamics. We further discussed the potential roles of less-studied phenomena: hippocampal SWRs, memory replay and neuromodulatory activity in the noradrenergic and serotonergic systems. We surveyed how this knowledge may advance our understanding of mental health and lead to clinical applications. In this final section, we highlight a few methodological directions that may clarify mechanisms, lead to reliable biomarkers and facilitate the development of personalized therapies toward the goals of better understanding and improving mental health.

Studying spontaneous and off-task thoughts and their neural mechanisms presents some unique challenges. Most standard experimental models rely on external perceptual or task-based manipulations that cannot typically be used to elicit and precisely measure these thoughts (see, however, refs. 82–84 for examples of how such manipulations can indirectly influence involuntary thoughts).

The introduction of experience sampling as a novel technique has led to important advances, and additional new approaches could yield further improvements in our ability to study thought dynamics. Recently, models involving free-flowing descriptions of thoughts have been applied to characterize the dynamic trajectories of conceptual associations^{134,192–194}. Moreover, the way that thoughts change over time can be altered in conditions that affect mental health. Semantic network analyses revealed that individuals high in rumination frequently became ‘stuck’ on negative, self-referential topics¹³⁴. The recent integration of descriptive experience-sampling¹⁹⁵, free-association⁸⁰ and think-aloud⁸¹ paradigms with fMRI highlight the potential of these approaches to offer further insights into altered neural and thought dynamics that affect mental health.

Innovations are also needed to improve the efficiency of discovering brain-experience relationships. Online experience-sampling neuroimaging studies treat experience and brain activity, respectively, as independent and dependent variables. In an alternative, brain-triggered experience-sampling model, neural activity is analysed in real time, and experience-sampling moments are initiated when a pre-specified event of interest (for example, SWR or DMN activation) occurs. Compared with the standard approach of sampling experience at random intervals, this approach could enhance the efficiency because sampling occurs selectively when a pre-hypothesized phenomenological experience is arising⁶⁹. In one application, real-time analysis detected a pre-specified EEG pattern that was predictive of dream contents¹⁹⁶. It was also suggested that real-time analyses could shed light on whether or not spontaneous memory reactivations are consciously experienced¹⁹⁷. In clinical applications that use experience sampling to identify neuro-modulation targets, brain-triggered experience sampling could reduce the time that a patient needs to be monitored (for example, ten days of iEEG recording was described in a recent case report⁹³).

Finally, it is critical to note that research on the neural basis of spontaneous and off-task thought has focused almost exclusively on groups of individuals. Brain mechanisms that give rise to self-labeled experiences may vary across and even within an individual across different contexts. Thus, personally derived neural markers may enable researchers to test, rather than assume, generalizability across individuals and instances. This issue is critical in the context of mental illness, where thought phenomenology can deviate substantially from the typical pattern. Recent findings have illustrated that idiographic, rather than group-level, fMRI predictive models are needed to predict affective valence when spontaneous thoughts have high self-relevance. Moreover, dense, individual-level, online experience sampling with fMRI recently revealed individual-specific neural patterns predictive of spontaneous sensory imagery⁸⁷. Building on these findings, future studies may derive personalized biomarkers of spontaneous and off-task thought that could aid in guiding brain-based therapies (for example, neurostimulation or neurofeedback).

These future directions highlight just a few ways in which research on spontaneous and off-task thought may be leveraged to guide assessments and treatments for mental health. These directions also highlight that the field is still in its infancy, and as such there remains enormous potential for research and clinical practice.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The dataset that was generated for Fig. 1 is available at <https://github.com/DynamicBrainMind/NatMentHealth2023>.

Code availability

The code that was used to generate Fig. 1 is available at <https://github.com/DynamicBrainMind/NatMentHealth2023>.

References

- Kane, M. J. et al. For whom the mind wanders, and when: an experience-sampling study of working memory and executive control in daily life. *Psychol. Sci.* **18**, 614–621 (2007).
- Killingsworth, M. A. & Gilbert, D. T. A wandering mind is an unhappy mind. *Science* **330**, 932–932 (2010).
- Klinger, E. & Cox, W. M. Dimensions of thought flow in everyday life. *Imagin. Cogn. Pers.* **7**, 105–128 (1987).
- Kane, M. J. et al. For whom the mind wanders, and when, varies across laboratory and daily-life settings. *Psychol. Sci.* **28**, 1271–1289 (2017).
- Seli, P. et al. How pervasive is mind wandering, really? *Conscious. Cogn.* **66**, 74–78 (2018).
- Baird, B. et al. Inspired by distraction: mind wandering facilitates creative incubation. *Psychol. Sci.* **23**, 1117–1122 (2012).
- Baird, B., Smallwood, J. & Schooler, J. W. Back to the future: autobiographical planning and the functionality of mind-wandering. *Conscious. Cogn.* **20**, 1604–1611 (2011).
- Smallwood, J. & Andrews-Hanna, J. Not all minds that wander are lost: the importance of a balanced perspective on the mind-wandering state. *Front. Psychol.* **4**, 441 (2013).
- Schafer, K. M., Lieberman, A., Sever, A. C. & Joiner, T. Prevalence rates of anxiety, depressive, and eating pathology symptoms between the pre- and peri-COVID-19 eras: a meta-analysis. *J. Affect. Disord.* **298**, 364–372 (2022).
- Zhdanova, M. et al. The prevalence and national burden of treatment-resistant depression and major depressive disorder in the United States. *J. Clin. Psychiatry* **82**, 20m13699 (2021).
- Bozhilova, N. S., Michelini, G., Kuntsi, J. & Asherson, P. Mind wandering perspective on attention-deficit/hyperactivity disorder. *Neurosci. Biobehav. Rev.* **92**, 464–476 (2018).
- Chaieb, L., Hoppe, C. & Fell, J. Mind wandering and depression: a status report. *Neurosci. Biobehav. Rev.* **133**, 104505 (2022).
- Marchetti, I., Koster, E. H. W., Klinger, E. & Alloy, L. B. Spontaneous thought and vulnerability to mood disorders: the dark side of the wandering mind. *Clin. Psychol. Sci.* **4**, 835–857 (2016).
- Christoff, K., Irving, Z. C., Fox, K. C. R., Spreng, R. N. & Andrews-Hanna, J. R. Mind-wandering as spontaneous thought: a dynamic framework. *Nat. Rev. Neurosci.* **17**, 718–731 (2016).
- Kucyi, A. Just a thought: how mind-wandering is represented in dynamic brain connectivity. *Neuroimage* **180**, 505–514 (2018).
- O’Callaghan, C., Walpola, I. C. & Shine, J. M. Neuromodulation of the mind-wandering brain state: the interaction between neuromodulatory tone, sharp wave-ripples and spontaneous thought. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **376**, 20190699 (2021).
- Smallwood, J. et al. The neural correlates of ongoing conscious thought. *iScience* **24**, 102132 (2021).
- Mittner, M., Hawkins, G. E., Boekel, W. & Forstmann, B. U. A neural model of mind wandering. *Trends Cogn. Sci.* **20**, 570–578 (2016).
- Girn, M., Mills, C., Roseman, L., Carhart-Harris, R. L. & Christoff, K. Updating the dynamic framework of thought: creativity and psychedelics. *Neuroimage* **213**, 116726 (2020).
- Kam, J. W. Y., Mittner, M. & Knight, R. T. Mind-wandering: mechanistic insights from lesion, tDCS, and iEEG. *Trends Cogn. Sci.* **26**, 268–282 (2022).
- Mildner, J. N. & Tamir, D. I. Spontaneous thought as an unconscious memory process. *Trends Neurosci.* **42**, 763–777 (2019).
- Fox, K. C. R., Andrews-Hanna, J. R. & Christoff, K. The neurobiology of self-generated thought from cells to systems: Integrating evidence from lesion studies, human intracranial electrophysiology, neurochemistry, and neuroendocrinology. *Neuroscience* **335**, 134–150 (2016).
- Fox, K. C. R. & Christoff, K. (eds) *The Oxford Handbook of Spontaneous Thought: Mind-wandering, Creativity, and Dreaming* (Oxford Univ. Press, 2018).

24. DuPre, E. & Spreng, R. N. in *The Oxford Handbook of Spontaneous Thought* (eds Fox, K. C. R. & Christoff, K.) 509–520 (Oxford Univ. Press, 2018).
25. Seli, P. et al. Mind-wandering as a natural kind: a family-resemblances view. *Trends Cogn. Sci.* **22**, 479–490 (2018).
26. Metzinger, T. in *The Oxford Handbook of Spontaneous Thought* (eds Fox, K. C. R. & Christoff, K.) 97–112 (Oxford Univ. Press, 2018).
27. Maillet, D. & Schacter, D. L. When the mind wanders: distinguishing stimulus-dependent from stimulus-independent thoughts during incidental encoding in young and older adults. *Psychol. Aging* **31**, 370–379 (2016).
28. Babo-Rebelo, M., Richter, C. G. & Tallon-Baudry, C. Neural responses to heartbeats in the default network encode the self in spontaneous thoughts. *J. Neurosci.* **36**, 7829–7840 (2016).
29. Avitan, L. & Stringer, C. Not so spontaneous: multi-dimensional representations of behaviors and context in sensory areas. *Neuron* **110**, 3064–3075 (2022).
30. Smallwood, J. Distinguishing how from why the mind wanders: a process-occurrence framework for self-generated mental activity. *Psychol. Bull.* **139**, 519–535 (2013).
31. Mills, C., Raffaelli, Q., Irving, Z. C., Stan, D. & Christoff, K. Is an off-task mind a freely-moving mind? Examining the relationship between different dimensions of thought. *Conscious. Cogn.* **58**, 20–33 (2018).
32. Mills, C., Porter, A. R., Andrews-Hanna, J. R., Christoff, K. & Colby, A. How task-unrelated and freely moving thought relate to affect: evidence for dissociable patterns in everyday life. *Emotion* **21**, 1029–1040 (2021).
33. Turnbull, A. et al. Reductions in task positive neural systems occur with the passage of time and are associated with changes in ongoing thought. *Sci. Rep.* **10**, 9912 (2020).
34. Christoff, K. et al. Mind-wandering as a scientific concept: cutting through the definitional haze. *Trends Cogn. Sci.* **22**, 957–959 (2018).
35. McVay, J. C. & Kane, M. J. Does mind wandering reflect executive function or executive failure? Comment on Smallwood and Schooler (2006) and Watkins (2008). *Psychol. Bull.* **136**, 188–197 (2010).
36. Mace, J. H. Involuntary autobiographical memory chains: implications for autobiographical memory organization. *Front. Psychiatry* **5**, 183 (2014).
37. McVay, J. C. & Kane, M. J. Conducting the train of thought: working memory capacity, goal neglect, and mind wandering in an executive-control task. *J. Exp. Psychol. Learn. Mem. Cogn.* **35**, 196–204 (2009).
38. Stawarczyk, D., Majerus, S., Maj, M., Van der Linden, M. & D'Argembeau, A. Mind-wandering: phenomenology and function as assessed with a novel experience sampling method. *Acta Psychol.* **136**, 370–381 (2011).
39. Smallwood, J. M., Baracaia, S. F., Lowe, M. & Obonsawin, M. Task unrelated thought whilst encoding information. *Conscious. Cogn.* **12**, 452–484 (2003).
40. Smallwood, J., Fishman, D. J. & Schooler, J. W. Counting the cost of an absent mind: mind wandering as an underrecognized influence on educational performance. *Psychon. Bull. Rev.* **14**, 230–236 (2007).
41. Smallwood, J. Mind-wandering while reading: attentional decoupling, mindless reading and the cascade model of inattention. *Lang. Linguist. Compass* **5**, 63–77 (2011).
42. Galéra, C. et al. Mind wandering and driving: responsibility case-control study. *Br. Med. J.* **345**, e8105 (2012).
43. Robison, M. K., Gath, K. I. & Unsworth, N. The neurotic wandering mind: an individual differences investigation of neuroticism, mind-wandering, and executive control. *Q. J. Exp. Psychol. (Hove)* **70**, 649–663 (2017).
44. Seli, P., Risko, E. F., Purdon, C. & Smilek, D. Intrusive thoughts: linking spontaneous mind wandering and OCD symptomatology. *Psychol. Res.* **2**, 392–398 (2016).
45. Smallwood, J., Fitzgerald, A., Miles, L. K. & Phillips, L. H. Shifting moods, wandering minds: negative moods lead the mind to wander. *Emotion* **9**, 271–276 (2009).
46. Engert, V., Smallwood, J. & Singer, T. Mind your thoughts: associations between self-generated thoughts and stress-induced and baseline levels of cortisol and alpha-amylase. *Biol. Psychol.* **103**, 283–291 (2014).
47. Cárdenas-Egúisquiza, A. L. & Berntsen, D. Sleeping poorly is robustly associated with a tendency to engage in spontaneous waking thought. *Conscious. Cogn.* **105**, 103401 (2022).
48. Hoffmann, F., Banzhaf, C., Kanske, P., BERPohl, F. & Singer, T. Where the depressed mind wanders: self-generated thought patterns as assessed through experience sampling as a state marker of depression. *J. Affect. Disord.* **198**, 127–134 (2016).
49. Smith, A. C., Brosowsky, N. P., Caron, E. E., Seli, P. & Smilek, D. Examining the relation between mind wandering and unhealthy eating behaviours. *Pers. Individ. Dif.* **200**, 111908 (2023).
50. Seli, P., Smallwood, J., Cheyne, J. A. & Smilek, D. On the relation of mind wandering and ADHD symptomatology. *Psychon. Bull. Rev.* **22**, 629–636 (2015).
51. Vatansever, D., Bozhilova, N. S., Asherson, P. & Smallwood, J. The devil is in the detail: exploring the intrinsic neural mechanisms that link attention-deficit/hyperactivity disorder symptomatology to ongoing cognition. *Psychol. Med.* **49**, 1185–1194 (2019).
52. Fox, K. C. R. et al. Affective neuroscience of self-generated thought. *Ann. N. Y. Acad. Sci.* **1426**, 25–51 (2018).
53. Franklin, M. S. et al. The silver lining of a mind in the clouds: interesting musings are associated with positive mood while mind-wandering. *Front. Psychol.* **4**, 583 (2013).
54. Welz, A., Reinhard, I., Alpers, G. W. & Kuehner, C. Happy thoughts: mind wandering affects mood in daily life. *Mindfulness* **9**, 332–343 (2018).
55. Smallwood, J. & O'Connor, R. C. Imprisoned by the past: unhappy moods lead to a retrospective bias to mind wandering. *Cogn. Emot.* **25**, 1481–1490 (2011).
56. Ruby, F. J. M., Smallwood, J., Engen, H. & Singer, T. How self-generated thought shapes mood—the relation between mind-wandering and mood depends on the socio-temporal content of thoughts. *PLoS ONE* **8**, e77554 (2013).
57. Webb, C. A. et al. Spontaneous thought characteristics are differentially related to heightened negative affect versus blunted positive affect in adolescents: an experience sampling study. *JCPP Adv.* **2**, e12110 (2022).
58. Stawarczyk, D. in *The Oxford Handbook of Spontaneous Thought* (eds Fox, K. C. R. & Christoff, K.) 193–214 (Oxford Univ. Press, 2018).
59. Mildner, J. N. & Tamir, D. I. The people around you are inside your head: social context shapes spontaneous thought. *J. Exp. Psychol. Gen.* **150**, 2375–2386 (2021).
60. Ji, J. L., Holmes, E. A., MacLeod, C. & Murphy, F. C. Spontaneous cognition in dysphoria: reduced positive bias in imagining the future. *Psychol. Res.* **83**, 817–831 (2019).
61. Tulving, E. Multiple memory systems and consciousness. *Hum. Neurobiol.* **6**, 67–80 (1987).
62. Gable, S. L., Hopper, E. A. & Schooler, J. W. When the muses strike: creative ideas of physicists and writers routinely occur during mind wandering. *Psychol. Sci.* **30**, 396–404 (2019).
63. Thiemann, R. F., Mills, C. & Kam, J. W. Y. Differential relationships between thought dimensions and momentary affect in daily life. *Psychol. Res.* <https://doi.org/10.1007/s00426-022-01766-9> (2022).
64. Turnbull, A. et al. Left dorsolateral prefrontal cortex supports context-dependent prioritisation of off-task thought. *Nat. Commun.* **10**, 3816 (2019).

65. Ho, N. S. P. et al. Facing up to the wandering mind: patterns of off-task laboratory thought are associated with stronger neural recruitment of right fusiform cortex while processing facial stimuli. *Neuroimage* **214**, 116765 (2020).
66. Gusnard, D. A., Raichle, M. E. & Raichle, M. E. Searching for a baseline: functional imaging and the resting human brain. *Nat. Rev. Neurosci.* **2**, 685–694 (2001).
67. Ingvar, D. H. 'Hyperfrontal' distribution of the cerebral grey matter flow in resting wakefulness; on the functional anatomy of the conscious state. *Acta Neurol. Scand.* **60**, 12–25 (1979).
68. Andreasen, N. C. et al. Remembering the past: two facets of episodic memory explored with positron emission tomography. *Am. J. Psychiatry* **152**, 1576–1585 (1995).
69. Gonzalez-Castillo, J., Kam, J. W. Y., Hoy, C. W. & Bandettini, P. A. How to interpret resting-state fMRI: ask your participants. *J. Neurosci.* **41**, 1130–1141 (2021).
70. McGuire, P. K., Paulesu, E., Frackowiak, R. S. & Frith, C. D. Brain activity during stimulus independent thought. *Neuroreport* **7**, 2095–2099 (1996).
71. Binder, J. R. et al. Conceptual processing during the conscious resting state. A functional MRI study. *J. Cogn. Neurosci.* **11**, 80–95 (1999).
72. Mason, M. F. et al. Wandering minds: the default network and stimulus-independent thought. *Science* **315**, 393–395 (2007).
73. Christoff, K., Gordon, A. M., Smallwood, J., Smith, R. & Schooler, J. W. Experience sampling during fMRI reveals default network and executive system contributions to mind wandering. *Proc. Natl Acad. Sci. USA* **106**, 8719–8724 (2009).
74. Lehmann, D., Henggeler, B., Koukkou, M. & Michel, C. M. Source localization of brain electric field frequency bands during conscious, spontaneous, visual imagery and abstract thought. *Cogn. Brain Res.* **1**, 203–210 (1993).
75. Weinstein, Y., De Lima, H. J. & van der Zee, T. Are you mind-wandering, or is your mind on task? The effect of probe framing on mind-wandering reports. *Psychon. Bull. Rev.* **25**, 754–760 (2018).
76. Jordão, M., Ferreira-Santos, F., Pinho, M. S. & St Jacques, P. L. Meta-analysis of aging effects in mind wandering: methodological and sociodemographic factors. *Psychol. Aging* **34**, 531–544 (2019).
77. Martinon, L. M., Smallwood, J., McGann, D., Hamilton, C. & Riby, L. M. The disentanglement of the neural and experiential complexity of self-generated thoughts: a users guide to combining experience sampling with neuroimaging data. *Neuroimage* **192**, 15–25 (2019).
78. Ellamil, M. et al. Dynamics of neural recruitment surrounding the spontaneous arising of thoughts in experienced mindfulness practitioners. *Neuroimage* **136**, 186–196 (2016).
79. Hasenkamp, W., Wilson-Mendenhall, C. D., Duncan, E. & Barsalou, L. W. Mind wandering and attention during focused meditation: a fine-grained temporal analysis of fluctuating cognitive states. *Neuroimage* **59**, 750–760 (2012).
80. Kim, B., Andrews-Hanna, J. R., Han, J., Lee, E. & Woo, C.-W. When self comes to a wandering mind: brain representations and dynamics of self-generated concepts in spontaneous thought. *Sci. Adv.* **8**, eabn8616 (2022).
81. Li, H.-X. et al. Neural representations of self-generated thought during think-aloud fMRI. *Neuroimage* **265**, 119775 (2022).
82. Berntsen, D., Staugaard, S. R. & Sørensen, L. M. T. Why am I remembering this now? Predicting the occurrence of involuntary (spontaneous) episodic memories. *J. Exp. Psychol. Gen.* **142**, 426–444 (2013).
83. Hall, S. A., Brodar, K. E., LaBar, K. S., Berntsen, D. & Rubin, D. C. Neural responses to emotional involuntary memories in posttraumatic stress disorder: differences in timing and activity. *Neuroimage Clin.* **19**, 793–804 (2018).
84. Schlagman, S. & Kvavilashvili, L. Involuntary autobiographical memories in and outside the laboratory: how different are they from voluntary autobiographical memories? *Mem. Cognit.* **36**, 920–932 (2008).
85. Tusche, A., Smallwood, J., Bernhardt, B. C. & Singer, T. Classifying the wandering mind: revealing the affective content of thoughts during task-free rest periods. *Neuroimage* **97**, 107–116 (2014).
86. Sormaz, M. et al. Default mode network can support the level of detail in experience during active task states. *Proc. Natl Acad. Sci. USA* **115**, 9318–9323 (2018).
87. Hung, S.-M. & Hsieh, P.-J. Mind wandering in sensory cortices. *Neuroimage Rep.* **2**, 100073 (2022).
88. Kam, J. W. Y. et al. Distinct electrophysiological signatures of task-unrelated and dynamic thoughts. *Proc. Natl Acad. Sci. USA* **118**, e2011796118 (2021).
89. Andrillon, T., Burns, A., Mackay, T., Windt, J. & Tsuchiya, N. Predicting lapses of attention with sleep-like slow waves. *Nat. Commun.* **12**, 3657 (2021).
90. Mortaheb, S. et al. Mind blanking is a distinct mental state linked to a recurrent brain profile of globally positive connectivity during ongoing mentation. *Proc. Natl Acad. Sci. USA* **119**, e2200511119 (2022).
91. Bozhilova, N., Kuntsi, J., Rubia, K., Asherson, P. & Michelini, G. Event-related brain dynamics during mind wandering in attention-deficit/hyperactivity disorder: an experience-sampling approach. *Neuroimage Clin.* **35**, 103068 (2022).
92. Kucyi, A. et al. Prediction of stimulus-independent and task-unrelated thought from functional brain networks. *Nat. Commun.* **12**, 1793 (2021).
93. Scangos, K. W. et al. Closed-loop neuromodulation in an individual with treatment-resistant depression. *Nat. Med.* **27**, 1696–1700 (2021).
94. Tomescu, M. I. et al. Spontaneous thought and microstate activity modulation by social imitation. *Neuroimage* **249**, 118878 (2022).
95. Shulman, G. L. et al. Common blood flow changes across visual tasks: II. Decreases in cerebral cortex. *J. Cogn. Neurosci.* **9**, 648–663 (1997).
96. Raichle, M. E. et al. A default mode of brain function. *Proc. Natl Acad. Sci. USA* **98**, 676–682 (2001).
97. Greicius, M. D., Krasnow, B., Reiss, A. L. & Menon, V. Functional connectivity in the resting brain: a network analysis of the default mode hypothesis. *Proc. Natl Acad. Sci. USA* **100**, 253–258 (2003).
98. Andrews-Hanna, J. R., Smallwood, J. & Spreng, R. N. The default network and self-generated thought: component processes, dynamic control, and clinical relevance. *Ann. N. Y. Acad. Sci.* **1316**, 29–52 (2014).
99. Andrews-Hanna, J. R., Reidler, J. S., Huang, C. & Buckner, R. L. Evidence for the default network's role in spontaneous cognition. *J. Neurophysiol.* **104**, 322–335 (2010).
100. Fox, M. D. et al. The human brain is intrinsically organized into dynamic, anticorrelated functional networks. *Proc. Natl Acad. Sci. USA* **102**, 9673–9678 (2005).
101. Dixon, M. L. et al. Interactions between the default network and dorsal attention network vary across default subsystems, time, and cognitive states. *Neuroimage* **147**, 632–649 (2017).
102. Kucyi, A. et al. Electrophysiological dynamics of antagonistic brain networks reflect attentional fluctuations. *Nat. Commun.* **11**, 325 (2020).
103. Buckner, R. L., Andrews-Hanna, J. R. & Schacter, D. L. The brain's default network: anatomy, function, and relevance to disease. *Ann. N. Y. Acad. Sci.* **1124**, 1–38 (2008).
104. Stawarczyk, D., Majerus, S., Maquet, P. & D'Argembeau, A. Neural correlates of ongoing conscious experience: both task-unrelatedness and stimulus-independence are related to default network activity. *PLoS ONE* **6**, e16997 (2011).

105. Kucyi, A., Salomons, T. V. & Davis, K. D. Mind wandering away from pain dynamically engages antinociceptive and default mode brain networks. *Proc. Natl Acad. Sci. USA* **110**, 18692–18697 (2013).
106. Sonuga-Barke, E. J. S. & Castellanos, F. X. Spontaneous attentional fluctuations in impaired states and pathological conditions: a neurobiological hypothesis. *Neurosci. Biobehav. Rev.* **31**, 977–986 (2007).
107. Whitfield-Gabrieli, S. & Ford, J. M. Default mode network activity and connectivity in psychopathology. *Annu. Rev. Clin. Psychol.* **8**, 49–76 (2012).
108. Beaty, R. E., Benedek, M., Silvia, P. J. & Schacter, D. L. Creative cognition and brain network dynamics. *Trends Cogn. Sci.* **20**, 87–95 (2016).
109. Kucyi, A., Esterman, M., Riley, C. S. & Valera, E. M. Spontaneous default network activity reflects behavioral variability independent of mind-wandering. *Proc. Natl Acad. Sci. USA* **113**, 13899–13904 (2016).
110. Groot, J. M. et al. Probing the neural signature of mind wandering with simultaneous fMRI-EEG and pupillometry. *Neuroimage* **224**, 117412 (2021).
111. Murphy, C. et al. Distant from input: evidence of regions within the default mode network supporting perceptually-decoupled and conceptually-guided cognition. *Neuroimage* **171**, 393–401 (2018).
112. Konishi, M., McLaren, D. G., Engen, H. & Smallwood, J. Shaped by the past: the default mode network supports cognition that is independent of immediate perceptual input. *PLoS ONE* **10**, e0132209 (2015).
113. Gratton, C. et al. Functional brain networks are dominated by Stable group and individual factors, not cognitive or daily variation. *Neuron* **98**, 439–452.e5 (2018).
114. Harmelech, T. & Malach, R. Neurocognitive biases and the patterns of spontaneous correlations in the human cortex. *Trends Cogn. Sci.* **17**, 606–615 (2013).
115. Kucyi, A. & Davis, K. D. Dynamic functional connectivity of the default mode network tracks daydreaming. *Neuroimage* **100**, 471–480 (2014).
116. Godwin, C. A. et al. Functional connectivity within and between intrinsic brain networks correlates with trait mind wandering. *Neuropsychologia* **103**, 140–153 (2017).
117. Zhang, J. et al. What have we really learned from functional connectivity in clinical populations? *Neuroimage* **242**, 118466 (2021).
118. Liu, X. & Duyn, J. H. Time-varying functional network information extracted from brief instances of spontaneous brain activity. *Proc. Natl Acad. Sci. USA* **110**, 4392–4397 (2013).
119. Karapanagiotidis, T. et al. The psychological correlates of distinct neural states occurring during wakeful rest. *Sci. Rep.* **10**, 21121 (2020).
120. Mittner, M. et al. When the brain takes a break: a model-based analysis of mind wandering. *J. Neurosci.* **34**, 16286–16295 (2014).
121. Hutchison, R. M. et al. Dynamic functional connectivity: promise, issues, and interpretations. *Neuroimage* **80**, 360–378 (2013).
122. Martin, C. G., He, B. J. & Chang, C. State-related neural influences on fMRI connectivity estimation. *Neuroimage* **244**, 118590 (2021).
123. Andrews-Hanna, J. R., Reidler, J. S., Sepulcre, J., Poulin, R. & Buckner, R. L. Functional-anatomic fractionation of the brain's default network. *Neuron* **65**, 550–562 (2010).
124. Axelrod, V., Rees, G. & Bar, M. The default network and the combination of cognitive processes that mediate self-generated thought. *Nat. Hum. Behav.* **1**, 896–910 (2017).
125. Kernbach, J. M. et al. Subspecialization within default mode nodes characterized in 10,000 UK Biobank participants. *Proc. Natl Acad. Sci. USA* **115**, 12295–12300 (2018).
126. Andrews-Hanna, J. R. & Grilli, M. D. Mapping the imaginative mind: charting new paths forward. *Curr. Dir. Psychol. Sci.* **30**, 82–89 (2021).
127. Braga, R. M. & Buckner, R. L. Parallel interdigitated distributed networks within the individual estimated by intrinsic functional connectivity. *Neuron* **95**, 457–471.e5 (2017).
128. DiNicola, L. M., Braga, R. M. & Buckner, R. L. Parallel distributed networks dissociate episodic and social functions within the individual. *J. Neurophysiol.* **123**, 1144–1179 (2020).
129. Buckner, R. L. & DiNicola, L. M. The brain's default network: updated anatomy, physiology and evolving insights. *Nat. Rev. Neurosci.* **20**, 593–608 (2019).
130. Ranganath, C. & Ritchey, M. Two cortical systems for memory-guided behaviour. *Nat. Rev. Neurosci.* **13**, 713–726 (2012).
131. Konu, D. et al. A role for the ventromedial prefrontal cortex in self-generated episodic social cognition. *Neuroimage* **218**, 116977 (2020).
132. Ho, N. S. P. et al. Individual variation in patterns of task focused, and detailed, thought are uniquely associated within the architecture of the medial temporal lobe. *Neuroimage* **202**, 116045 (2019).
133. Smallwood, J. & Schooler, J. W. The science of mind wandering: empirically navigating the stream of consciousness. *Annu. Rev. Psychol.* **66**, 487–518 (2015).
134. Andrews-Hanna, J. R. et al. The conceptual building blocks of everyday thought: tracking the emergence and dynamics of ruminative and nonruminative thinking. *J. Exp. Psychol. Gen.* **151**, 628–642 (2022).
135. Kajimura, S., Kochiyama, T., Abe, N. & Nomura, M. Challenge to unity: relationship between hemispheric asymmetry of the default mode network and mind wandering. *Cereb. Cortex* **29**, 2061–2071 (2019).
136. Bertossi, E., Peccenini, L., Solmi, A., Avenanti, A. & Ciaramelli, E. Transcranial direct current stimulation of the medial prefrontal cortex dampens mind-wandering in men. *Sci. Rep.* **7**, 16962 (2017).
137. Filmer, H. L., Marcus, L. H. & Dux, P. E. Stimulating task unrelated thoughts: tDCS of prefrontal and parietal cortices leads to polarity specific increases in mind wandering. *Neuropsychologia* **151**, 107723 (2021).
138. Mowlem, F. D. et al. Validation of the mind excessively wandering scale and the relationship of mind wandering to impairment in adult ADHD. *J. Atten. Disord.* **23**, 624–634 (2019).
139. Bertossi, E. & Ciaramelli, E. Ventromedial prefrontal damage reduces mind-wandering and biases its temporal focus. *Soc. Cogn. Affect. Neurosci.* **11**, 1783–1791 (2016).
140. McCormick, C., Rosenthal, C. R., Miller, T. D. & Maguire, E. A. Mind-wandering in people with hippocampal damage. *J. Neurosci.* **38**, 2745–2754 (2018).
141. O'Callaghan, C., Shine, J. M., Hodges, J. R., Andrews-Hanna, J. R. & Irish, M. Hippocampal atrophy and intrinsic brain network dysfunction relate to alterations in mind wandering in neurodegeneration. *Proc. Natl Acad. Sci. USA* **116**, 3316–3321 (2019).
142. Chou, T., Hooley, J. M. & Camprodon, J. A. Transcranial direct current stimulation of default mode network parietal nodes decreases negative mind-wandering about the past. *Cognit. Ther. Res.* **44**, 10–20 (2020).
143. Dixon, M. L. et al. Heterogeneity within the frontoparietal control network and its relationship to the default and dorsal attention networks. *Proc. Natl Acad. Sci. USA* **115**, E1598–E1607 (2018).
144. Smallwood, J., Brown, K., Baird, B. & Schooler, J. W. Cooperation between the default mode network and the frontal-parietal network in the production of an internal train of thought. *Brain Res.* **1428**, 60–70 (2012).
145. Axelrod, V., Rees, G., Lavidor, M. & Bar, M. Increasing propensity to mind-wander with transcranial direct current stimulation. *Proc. Natl Acad. Sci. USA* **112**, 3314–3319 (2015).

146. Boayue, N. M. et al. The interplay between executive control, behavioural variability and mind wandering: Insights from a high-definition transcranial direct-current stimulation study. *Eur. J. Neurosci.* **53**, 1498–1516 (2021).
147. Uddin, L. Q. Salience processing and insular cortical function and dysfunction. *Nat. Rev. Neurosci.* **16**, 55–61 (2015).
148. Zerbi, V. et al. Rapid reconfiguration of the functional connectome after chemogenetic locus coeruleus activation. *Neuron* **103**, 702–718.e5 (2019).
149. Seeley, W. W. et al. Dissociable intrinsic connectivity networks for salience processing and executive control. *J. Neurosci.* **27**, 2349–2356 (2007).
150. Webb, C. A. et al. Mind-wandering in adolescents predicts worse affect and is linked to aberrant default mode network–salience network connectivity. *J. Am. Acad. Child Adolesc. Psychiatry* **60**, 377–387 (2021).
151. Lydon-Staley, D. M. et al. Repetitive negative thinking in daily life and functional connectivity among default mode, fronto-parietal, and salience networks. *Transl. Psychiatry* **9**, 234 (2019).
152. Schooler, J. W. et al. Meta-awareness, perceptual decoupling and the wandering mind. *Trends Cogn. Sci.* **15**, 319–326 (2011).
153. Zhang, M. et al. Perceptual coupling and decoupling of the default mode network during mind-wandering and reading. *Elife* **11**, e74011 (2022).
154. Woo, C.-W., Chang, L. J., Lindquist, M. A. & Wager, T. D. Building better biomarkers: brain models in translational neuroimaging. *Nat. Neurosci.* **20**, 365–377 (2017).
155. Shen, X. et al. Using connectome-based predictive modeling to predict individual behavior from brain connectivity. *Nat. Protoc.* **12**, 506–518 (2017).
156. Kirkby, L. A. et al. An amygdala–hippocampus subnetwork that encodes variation in human mood. *Cell* **175**, 1688–1700.e14 (2018).
157. Varela, F., Lachaux, J. P., Rodriguez, E. & Martinerie, J. The brainweb: phase synchronization and large-scale integration. *Nat. Rev. Neurosci.* **2**, 229–239 (2001).
158. Aru, J., Drüke, M., Pikamäe, J. & Larkum, M. E. Mental navigation and the neural mechanisms of insight. *Trends Neurosci.* **46**, 100–109 (2023).
159. Axmacher, N., Elger, C. E. & Fell, J. Ripples in the medial temporal lobe are relevant for human memory consolidation. *Brain* **131**, 1806–1817 (2008).
160. Norman, Y. et al. Hippocampal sharp-wave ripples linked to visual episodic recollection in humans. *Science* **365**, eaax1030 (2019).
161. Liu, Y., Dolan, R. J., Kurth-Nelson, Z. & Behrens, T. E. J. Human replay spontaneously reorganizes experience. *Cell* **178**, 640–652.e14 (2019).
162. Buzsáki, G. Hippocampal sharp wave-ripple: a cognitive biomarker for episodic memory and planning. *Hippocampus* **25**, 1073–1188 (2015).
163. Carr, M. F., Jadhav, S. P. & Frank, L. M. Hippocampal replay in the awake state: a potential substrate for memory consolidation and retrieval. *Nat. Neurosci.* **14**, 147–153 (2011).
164. Tambini, A. & Davachi, L. Persistence of hippocampal multivoxel patterns into postencoding rest is related to memory. *Proc. Natl Acad. Sci. USA* **110**, 19591–19596 (2013).
165. Norman, Y., Raccach, O., Liu, S., Parvizi, J. & Malach, R. Hippocampal ripples and their coordinated dialogue with the default mode network during recent and remote recollection. *Neuron* **109**, 2767–2780.e5 (2021).
166. Kaplan, R. et al. Hippocampal sharp-wave ripples influence selective activation of the default mode network. *Curr. Biol.* **26**, 686–691 (2016).
167. Higgins, C. et al. Replay bursts in humans coincide with activation of the default mode and parietal alpha networks. *Neuron* **109**, 882–893.e7 (2021).
168. Kaefer, K., Stella, F., McNaughton, B. L. & Battaglia, F. P. Replay, the default mode network and the cascaded memory systems model. *Nat. Rev. Neurosci.* **23**, 628–640 (2022).
169. Deeb, W. et al. Fornix-region deep brain stimulation-induced memory flashbacks in Alzheimer’s disease. *N. Engl. J. Med.* **381**, 783–785 (2019).
170. Nour, M. M., Liu, Y., Arumham, A., Kurth-Nelson, Z. & Dolan, R. J. Impaired neural replay of inferred relationships in schizophrenia. *Cell* **184**, 4315–4328.e17 (2021).
171. Heller, A. S. & Bagot, R. C. Is hippocampal replay a mechanism for anxiety and depression? *JAMA Psychiatry* **77**, 431–432 (2020).
172. McGinley, M. J. et al. Waking state: rapid variations modulate neural and behavioral responses. *Neuron* **87**, 1143–1161 (2015).
173. Aston-Jones, G. & Cohen, J. D. An integrative theory of locus coeruleus–norepinephrine function: adaptive gain and optimal performance. *Annu. Rev. Neurosci.* **28**, 403–450 (2005).
174. Joshi, S., Li, Y., Kalwani, R. M. & Gold, J. I. Relationships between pupil diameter and neuronal activity in the locus coeruleus, colliculi, and cingulate cortex. *Neuron* **89**, 221–234 (2016).
175. Reimer, J. et al. Pupil fluctuations track rapid changes in adrenergic and cholinergic activity in cortex. *Nat. Commun.* **7**, 13289 (2016).
176. Megemont, M., McBurney-Lin, J. & Yang, H. Pupil diameter is not an accurate real-time readout of locus coeruleus activity. *Elife* **11**, e70510 (2022).
177. Smallwood, J. et al. Pupillometric evidence for the decoupling of attention from perceptual input during offline thought. *PLoS ONE* **6**, e18298 (2011).
178. Unsworth, N. & Robison, M. K. Tracking arousal state and mind wandering with pupillometry. *Cogn. Affect. Behav. Neurosci.* **18**, 638–664 (2018).
179. Konishi, M., Brown, K., Battaglini, L. & Smallwood, J. When attention wanders: pupillometric signatures of fluctuations in external attention. *Cognition* **168**, 16–26 (2017).
180. Kucyi, A. & Parvizi, J. Pupillary dynamics link spontaneous and task-evoked activations recorded directly from human insula. *J. Neurosci.* **40**, 6207–6218 (2020).
181. Schindler, E. A. D. & D’Souza, D. C. The therapeutic potential of psychedelics. *Science* **378**, 1051–1053 (2022).
182. Madsen, M. K. et al. Psychedelic effects of psilocybin correlate with serotonin 2A receptor occupancy and plasma psilocin levels. *Neuropsychopharmacology* **44**, 1328–1334 (2019).
183. Spitzer, M. et al. Increased activation of indirect semantic associations under psilocybin. *Biol. Psychiatry* **39**, 1055–1057 (1996).
184. Carhart-Harris, R. L. et al. Neural correlates of the psychedelic state as determined by fMRI studies with psilocybin. *Proc. Natl Acad. Sci. USA* **109**, 2138–2143 (2012).
185. Kwan, A. C., Olson, D. E., Preller, K. H. & Roth, B. L. The neural basis of psychedelic action. *Nat. Neurosci.* **25**, 1407–1419 (2022).
186. Preller, K. H. et al. Changes in global and thalamic brain connectivity in LSD-induced altered states of consciousness are attributable to the 5-HT_{2A} receptor. *Elife* **7**, e35082 (2018).
187. Müller, F. et al. Increased thalamic resting-state connectivity as a core driver of LSD-induced hallucinations. *Acta Psychiatr. Scand.* **136**, 648–657 (2017).
188. Müller, F., Dolder, P. C., Schmidt, A., Liechti, M. E. & Borgwardt, S. Altered network hub connectivity after acute LSD administration. *Neuroimage Clin.* **18**, 694–701 (2018).
189. Mason, N. L. et al. Spontaneous and deliberate creative cognition during and after psilocybin exposure. *Transl. Psychiatry* **11**, 209 (2021).
190. Lebedev, A. V. et al. Finding the self by losing the self: neural correlates of ego-dissolution under psilocybin. *Hum. Brain Mapp.* **36**, 3137–3153 (2015).

191. Tagliazucchi, E. et al. Increased global functional connectivity correlates with LSD-induced ego dissolution. *Curr. Biol.* **26**, 1043–1050 (2016).
192. Raffaelli, Q. et al. The think aloud paradigm reveals differences in the content, dynamics and conceptual scope of resting state thought in trait brooding. *Sci. Rep.* **11**, 19362 (2021).
193. Bellana, B., Mahabal, A. & Honey, C. J. Narrative thinking lingers in spontaneous thought. *Nat. Commun.* **13**, 4585 (2022).
194. Li, H.-X. et al. Exploring self-generated thoughts in a resting state with natural language processing. *Behav. Res. Methods* **54**, 1725–1743 (2022).
195. Kühn, S., Fernyhough, C., Alderson-Day, B. & Hurlburt, R. T. Inner experience in the scanner: can high fidelity apprehensions of inner experience be integrated with fMRI? *Front. Psychol.* **5**, 1393 (2014).
196. Siclari, F. et al. The neural correlates of dreaming. *Nat. Neurosci.* **20**, 872–878 (2017).
197. Tambini, A. & Davachi, L. Awake reactivation of prior experiences consolidates memories and biases cognition. *Trends Cogn. Sci.* **23**, 876–890 (2019).
198. Van den Driessche, C. et al. Attentional lapses in attention-deficit/hyperactivity disorder: blank rather than wandering thoughts. *Psychol. Sci.* **28**, 1375–1386 (2017).
199. Watts, F. N., MacLeod, A. K. & Morris, L. Associations between phenomenal and objective aspects of concentration problems in depressed patients. *Br. J. Psychol.* **79**, 241–250 (1988).
200. Kawagoe, T., Onoda, K. & Yamaguchi, S. The neural correlates of ‘mind blanking’: when the mind goes away. *Hum. Brain Mapp.* **40**, 4934–4940 (2019).
201. Fell, J. What is mind blanking: a conceptual clarification. *Eur. J. Neurosci.* **56**, 4837–4842 (2022).
202. Borchers, S., Himmelbach, M., Logothetis, N. & Karnath, H.-O. Direct electrical stimulation of human cortex — the gold standard for mapping brain functions? *Nat. Rev. Neurosci.* **13**, 63–70 (2011).
203. Holtzheimer, P. E. & Mayberg, H. S. Deep brain stimulation for psychiatric disorders. *Annu. Rev. Neurosci.* **34**, 289–307 (2011).
204. Curot, J. et al. Memory scrutinized through electrical brain stimulation: a review of 80 years of experiential phenomena. *Neurosci. Biobehav. Rev.* **78**, 161–177 (2017).
205. Fox, K. C. R. et al. Intrinsic network architecture predicts the effects elicited by intracranial electrical stimulation of the human brain. *Nat. Hum. Behav.* **4**, 1039–1052 (2020).
206. Parvizi, J. et al. Altered sense of self during seizures in the postero-medial cortex. *Proc. Natl Acad. Sci. USA* **118**, e2100522118 (2021).
207. Shofty, B. et al. The default network is causally linked to creative thinking. *Mol. Psychiatry* **27**, 1848–1854 (2022).

Acknowledgements

This work was supported by the National Institute of Mental Health (NIMH) of the National Institutes of Health (NIH) under award numbers R21MH127384 (to A.K.), R21MH129630 (to A.K. and S.W.-G.) and R01MH125414 (J.R.A.-H.). J.W.Y.K. was supported by a Natural Sciences and Engineering Research Council (NSERC) discovery grant.

Author contributions

A.K. completed the initial draft. J.W.Y.K., J.R.A.-H., K.C. and S.W.-G. contributed to the writing, conceptualization and editing.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s44220-023-00133-w>.

Correspondence should be addressed to Aaron Kucyi.

Peer review information *Nature Mental Health* thanks Ioana Carcea, Jonathan Smallwood and the other, anonymous reviewer for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature America, Inc. 2023

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Please do not complete any field with "not applicable" or n/a. Refer to the help text for what text to use if an item is not relevant to your study. For final submission: please carefully check your responses for accuracy; you will not be able to make changes later.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☒	☐ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☐ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Terms used to search PubMed for Figure 1 are available at https://github.com/DynamicBrainMind/NatMentHealth2023 .
Data analysis	Code used to generate Figure 1 is available at https://github.com/DynamicBrainMind/NatMentHealth2023 .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The dataset that was generated for Figure 1 is available at <https://github.com/DynamicBrainMind/NatMentHealth2023>.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Reporting on race, ethnicity, or other socially relevant groupings

Population characteristics

Recruitment

Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences

☐ Behavioural & social sciences

☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Data exclusions

Replication

Randomization

Blinding

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description

Research sample

Sampling strategy

Data collection

Timing

Data exclusions

Non-participation

Randomization

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	
Research sample	
Sampling strategy	
Data collection	
Timing and spatial scale	
Data exclusions	
Reproducibility	
Randomization	
Blinding	

Did the study involve field work? ☐ Yes ☐ No

Field work, collection and transport

Field conditions	
Location	
Access & import/export	
Disturbance	

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	
Validation	

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

Authentication

Mycoplasma contamination

Commonly misidentified lines
(See [ICLAC](#) register)

Palaeontology and Archaeology

Specimen provenance

Specimen deposition

Dating methods

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

Wild animals

Reporting on sex

Field-collected samples

Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration

Study protocol

Data collection

Outcomes

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes
☒	☐ Public health
☒	☐ National security
☒	☐ Crops and/or livestock
☒	☐ Ecosystems
☒	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☒	☐ Demonstrate how to render a vaccine ineffective
☒	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☒	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☒	☐ Increase transmissibility of a pathogen
☒	☐ Alter the host range of a pathogen
☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☒	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks	
Novel plant genotypes	
Authentication	

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links <i>May remain private before publication.</i>	
Files in database submission	
Genome browser session (e.g. UCSC)	

Methodology

Replicates	
Sequencing depth	
Antibodies	
Peak calling parameters	
Data quality	
Software	

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Instrument

Software

Cell population abundance

Gating strategy

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Design specifications

Behavioral performance measures

Imaging type(s)

Field strength

Sequence & imaging parameters

Area of acquisition

Diffusion MRI

☐ Used

☐ Not used

Preprocessing

Preprocessing software

Normalization

Normalization template

Noise and artifact removal

Volume censoring

Statistical modeling & inference

Model type and settings

Effect(s) tested

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

(See [Eklund et al. 2016](#))

Correction

Models & analysis

n/a | Involved in the study

- | | | |
|--------------------------|--------------------------|--|
| ☐ | ☐ | Functional and/or effective connectivity |
| ☐ | ☐ | Graph analysis |
| ☐ | ☐ | Multivariate modeling or predictive analysis |

Functional and/or effective connectivity

Graph analysis

Multivariate modeling and predictive analysis

